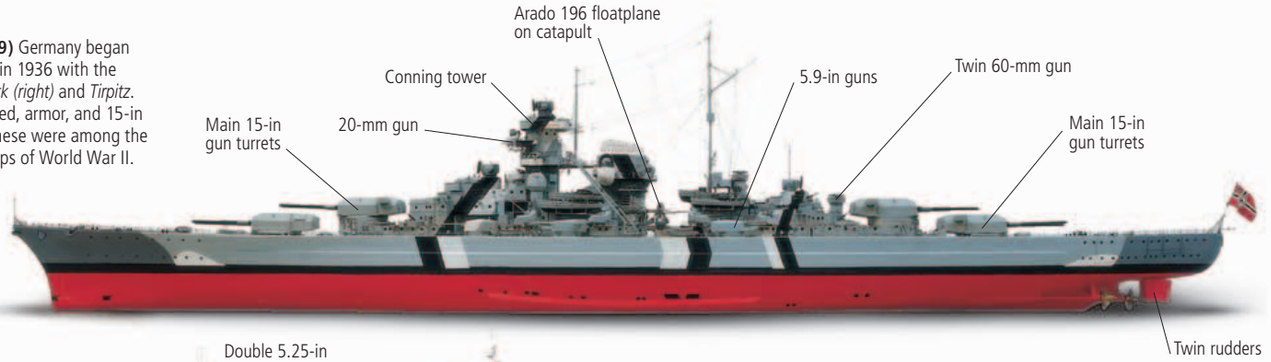
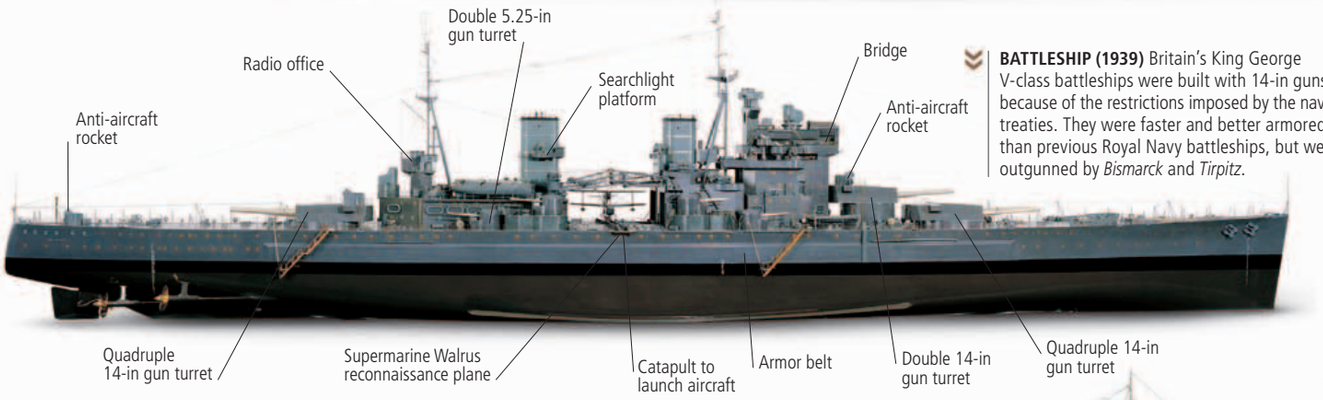
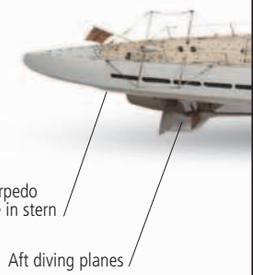
WORLD WAR II WARSHIPS

BATTLESHIP (1939) Germany began rebuilding its navy in 1936 with the battleships *Bismarck (right)* and *Tirpitz*. With their high speed, armor, and 15-in (380-mm) guns, these were among the most advanced ships of World War II.

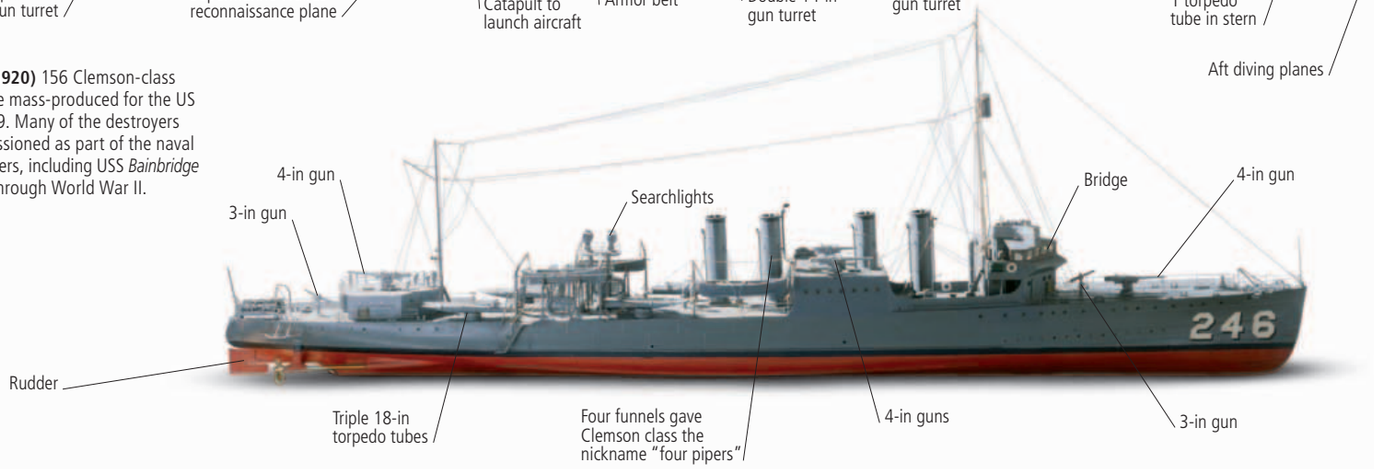


BATTLESHIP (1939) Britain's King George V-class battleships were built with 14-in guns because of the restrictions imposed by the naval treaties. They were faster and better armored than previous Royal Navy battleships, but were outgunned by *Bismarck* and *Tirpitz*.





DESTROYER (1920) 156 Clemson-class destroyers were mass-produced for the US Navy from 1919. Many of the destroyers were decommissioned as part of the naval treaties but others, including USS *Bainbridge* (*right*) served through World War II.



Major navies entered World War II still preoccupied with developing bigger and better battleships. Ships under construction when the war started included the German *Tirpitz* and *Bismarck*, the American *Iowa*, and the Japanese *Yamato* and *Musashi*, the two largest battleships ever built. Capable of speeds of around 30 knots and with a main armament of 15-in to 18.1-in guns, these capital ships were extraordinary fighting machines. But although big-gun ships played an important part in the war, they were upstaged by the rise of air power.

Warships were fitted with ever greater numbers of anti-aircraft guns, but proved painfully vulnerable to land- and carrier-based air attack. Aircraft carriers had been under development since World War I, and during World War II naval air power came to maturity. Equipped with a mix of dive-bombers, torpedo, and fighter aircraft, fleet carriers proved a dominant force especially in the Pacific theater, where some naval battles were fought exclusively between carriers. Smaller escort carriers provided air cover for convoys.



Explosive paravane
This underwater glider was an anti-submarine device which was towed behind a ship. It exploded on impact with a submarine.

SUBMARINE WARFARE

Powered by diesel engines on the surface and electric motors under the sea, submarines remained relatively primitive "submersibles" until relatively late in the war, when the Germans introduced U-boats capable of sustained underwater operations. Yet even these unsophisticated boats had a critical impact in both the Pacific and the Atlantic theaters. A whole new range of anti-submarine weaponry was deployed aboard destroyers, frigates, sloops, and corvettes. Radar and sonar detection technologies became integral to naval warfare, particularly in detecting enemy aircraft or submarines. The importance of amphibious operations in World War II also drove the development of a broad spectrum of landing craft.